

Spend the Wasted Epochs: Preference Post-Training from the Saturation Checkpoint for Sample-Efficient Language Modeling

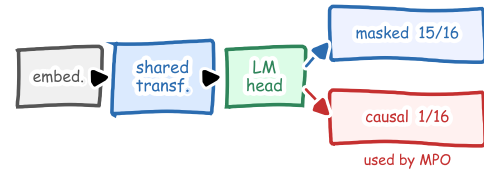
Anonymous ACL submission

Abstract

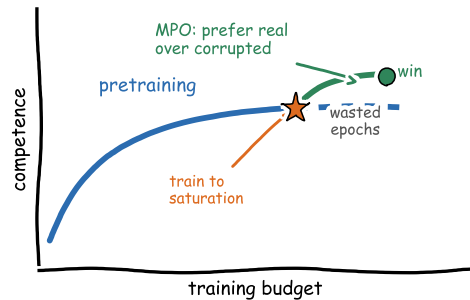
Sample-efficient pretraining under a developmental data budget, such as the ten million words and ten epochs of the BabyLM Strict-Small track, is usually approached by adding training signal: a new objective, optimizer, curriculum, or teacher. The budget itself receives less scrutiny. We show that much of it is spent after learning has stopped. A 33.0M-parameter GPT-BERT hybrid trained under the track rules saturates on downstream evaluations by roughly 70M words seen, about 70% of its permitted exposure, while training loss continues to fall; the final epochs improve fit, not competence. From this diagnosis we derive three design principles, timing, preservation, and alignment, and instantiate them as *Minimal-Pair Preference Optimization* (MPO): a short post-training phase, begun at the saturation checkpoint, that teaches the model to prefer real corpus sentences over length-preserving corruptions of them under a frozen-reference penalty. A controlled study of seven in-training interventions supports the principles: none improves on the baseline, and the unanchored ones damage the same cluster of long-distance dependencies first. MPO improves five of the seven official task averages, concedes under a point on the other two, is alone in repairing that cluster, and matches the fully trained baseline with 7.2 rather than 10 epochs; our MPO entry places fourth in the track.

1 Introduction

A child hears no more than about 100 million words by early adolescence. A language model consumes several orders of magnitude more before it is called trained. The BabyLM Challenge asks what this gap conceals: how much linguistic competence can pretraining extract from a corpus a child could plausibly have heard? (Warstadt et al., 2023; Charpentier et al., 2025) We compete in the Strict-Small track of the 2026 edition (Choshen et al., 2026), the



(a) Hybrid backbone



(b) Reallocating the wasted budget

Figure 1: **The idea in one picture.** (a) The hybrid backbone: one shared transformer and a single language-model head, trained on both a masked objective (15/16 of batches) and a causal one (1/16). The causal mode is what MPO later exploits. (b) Standard pretraining saturates, then wastes its final epochs (dashed). Starting from that saturation checkpoint, MPO spends the left-over budget teaching the model to prefer real sentences over corrupted ones, and matches the fully-trained baseline in fewer epochs.

tightest form of the question: at most 10 million words of text, with the corpus traversed at most ten times.

Most entries respond by adding signal. New objectives, curricula, architectures, tokenizers, and distilled teachers arrive every year (Hu et al., 2024; Charpentier et al., 2025). We start from a different question: before adding anything, when does the existing budget do its work? Training a 33.0M-parameter GPT-BERT hybrid (Charpentier and Samuel, 2024) under the track rules, we find that downstream competence saturates by roughly

70M words seen, seven of the ten permitted epochs, while training loss keeps falling. Those last epochs purchase a lower loss, not new competence. A per-phenomenon breakdown adds a second fact: a small cluster of long-distance dependencies is acquired last and stays fragile to the end.

The diagnosis dictates what a useful intervention must look like. It should arrive after saturation, where its gradient no longer competes with language modeling for capacity. It must be anchored, because fragile structure is what strong new gradients destroy first. And it should share the functional form of the evaluation, which for the core benchmarks is a likelihood comparison between minimally different strings. We call these the timing, preservation, and alignment principles.

Their instantiation is Minimal-Pair Preference Optimization (MPO): a short preference-optimization phase (Rafailov et al., 2023) that restarts from the saturation checkpoint and teaches the model to prefer real corpus sentences over length-preserving corruptions of them, with a frozen reference penalizing drift (Figure 1). The principles are tested rather than assumed. A controlled study of seven in-training interventions drawn from BabyLM entries and the broader pre-training literature confirms both predictions: none improves on the baseline, and the unanchored ones damage the fragile cluster first. MPO improves five of the seven official task columns and concedes under a point on the rest, is alone in repairing the fragile cluster, and matches the fully trained baseline after 7.2 rather than 10 epochs of exposure; our MPO entry places fourth on the track leaderboard, the fully trained baseline eighth.

Beyond the method, the paper contributes the saturation diagnosis, cheap to run and decisive about where post-training signal belongs, and the controlled study of why published gains fail to transfer here.

2 Related Work

The BabyLM Challenge has run three times, and its findings reports serve as a census of what helps under developmental budgets (Warstadt et al., 2023; Hu et al., 2024; Charpentier et al., 2025). The recipes that succeed tend to change what the model predicts, not when. GPT-BERT, our backbone, won the 2024 edition by blending masked and causal prediction in a single transformer (Charpentier and Samuel, 2024). In the 2025 edition, reweighting

the masking distribution as training progresses outperformed the official strict-small baseline (Edman and Fraser, 2025). BabyLlama distilled an ensemble of teachers into one small student and later served as an official baseline (Timiryasov and Tastet, 2023; Hu et al., 2024). All of these run as a single uninterrupted phase; where in the budget the signal lands is not itself a design variable. That variable is our subject.

Timing has been studied descriptively: Chang and Bergen (2022) chart when individual words are acquired during pretraining, regularly enough to compare against child norms, and the 2026 evaluation includes an age-of-acquisition task built on this lens (Choshen et al., 2026). We push the lens from describing when learning happens to acting on when it stops.

The interventions in our controlled study (§4.1) each arrive with a published pedigree: span corruption (Joshi et al., 2020), selective token losses that concentrate compute on learnable, worth-learning points (Mindermann et al., 2022; Lin et al., 2024), the Muon optimizer (Jordan et al., 2024), weight averaging in its stochastic and soup forms (Izmailov et al., 2018; Wortsman et al., 2022), and morpheme-aware segmentation from a child-inspired entry (Bölücü and Can, 2025). The reported gains come from other regimes: larger models, longer budgets, a previous edition’s setting. Whether they hold under a 10M-word budget is empirical, and §4.1’s answer is that none does. What informs our method is not the failures but the pattern in how they fail.

Preference optimization entered the toolkit as an alignment method: DPO tunes a model to prefer one completion over another under a KL penalty toward a frozen reference, with pairs elicited from human judgment (Rafailov et al., 2023). Its one BabyLM appearance so far targets dialogue: Padovani et al. (2025) tune a dialogue-pretrained model with DPO toward communicative continuations, gaining on a custom dialogue benchmark while conceding the standard suite. MPO retains the objective and replaces everything around it. Pairs are a corpus sentence and a synthetic corruption of it, so no annotator, reward model, or teacher is involved; the target is grammatical competence, not helpfulness; and the phase is placed by measurement at the saturation checkpoint, not appended to a finished model. The corruption design itself borrows from minimal-pair evaluation, in which BLiMP isolates one phenomenon per item by contrasting two strings that differ minimally

(Warstadt et al., 2020). Learning from corruptions also has a pretraining pedigree: ELECTRA trains a token-level discriminator against sampled replacements throughout pretraining (Clark et al., 2020), where MPO compares whole sequences, after acquisition, under an anchor. We train on the form of comparison the benchmarks measure, computed entirely on text the rules already allow.

3 Method

3.1 The Hybrid Backbone and Training Setup

We train on the official Strict-Small corpus: 10M words across six sources, dominated by transcribed speech (CHILDES 28%, OpenSubtitles 23%, BNC spoken 8%, Switchboard) alongside Project Gutenberg (26%) and Simple Wikipedia (15%). A byte-level BPE tokenizer with vocabulary 16,384 is fit on the training split alone, and documents are packed into a single token stream read at sequence length 128.

The backbone is a GPT-BERT hybrid (Charpentier and Samuel, 2024): masked and causal prediction on a single transformer with one language-model head and shifted next-position targets for both modes,

$$\mathcal{L}(\theta) = \mathbb{E}_B [z \text{CE}_{\text{mask}}(B; \theta) + (1-z) \text{CE}_{\text{causal}}(B; \theta)], \quad (1)$$

with $z \sim \text{Bernoulli}(15/16)$ per micro-batch. The model has 33,046,084 parameters (hidden 384, 12 layers, 6 heads, tied embeddings) and is trained for 10 epochs ($\approx 99M$ words seen) with AdamW, a cosine schedule, and a masking rate annealed from 0.30 to 0.15, on a single consumer laptop (M1, 16 GB, MPS). Appendix A lists every hyperparameter; none was tuned against the official evaluation.

Evaluation is the organizers’ 2026 pipeline, unmodified. The zero-shot tasks compare the model’s likelihoods over candidate strings: BLiMP (Warstadt et al., 2020) with its challenge supplement (Warstadt et al., 2023), EWoK (Ivanova et al., 2024), COMPS (Misra et al., 2023), entity tracking (Kim and Schuster, 2023), and GlobalPIQA (Chang et al., 2025). The suite adds reading-time prediction, a word age-of-acquisition comparison against child norms following Chang and Bergen (2022), and a (Super)GLUE finetuning battery (Wang et al., 2018, 2019). During development we gate decisions with the pipeline’s fast subsets (200 BLiMP and 50 supplement items per subtask) and reserve the full server-scored suite for the submitted models (§4.3).

3.2 Diagnosing the Budget

Before proposing an intervention, we ask a prior question: over the course of training, when does the model stop *learning* as opposed to continuing to *fit*? Two observations about the baseline hybrid answer it, and they shape every design choice that follows.

The first observation concerns timing: evaluation saturates well before the budget is spent. Figure 2 plots fast BLiMP accuracy against words seen, overlaid with the masked training loss. The two curves dissociate over the final third of training. Accuracy rises steeply from 54.21 at 1M words to 68.74 at 50M and then goes flat: 70.13 at 70M words, 70.01 at 100M, a spread inside the seed-noise band of §4.2. Writing $A(w)$ for accuracy after w words seen, define the saturation point as the earliest milestone that no later milestone beats by more than a tolerance ε ,

$$w^* = \min \left\{ w : \max_{w' > w} A(w') - A(w) \leq \varepsilon \right\}. \quad (2)$$

On this trajectory $w^* = 70M$ for every ε below 1.2 points, seventeen times the seed band, so nothing about the choice of checkpoint hinges on the tolerance (Appendix C sweeps it). The training loss, in contrast, continues its steady descent over exactly this span (2.71 at 40M $\rightarrow$ 2.19 at 98M). The last $\sim 30\%$ of the epoch budget therefore yields a lower training loss but no downstream competence: those epochs memorize a corpus already seen seven times rather than generalizing from it.

The second observation concerns what is fragile. A per-phenomenon breakdown (Appendix B) shows morphology mastered early and world knowledge near chance throughout, but singles out a small cluster of long-distance dependencies, filler-gap licensing, distractor agreement across a relational noun, and N -bar ellipsis, acquired last and the most fragile; we call this cluster $\mathcal{F}$. As §4.1 shows, $\mathcal{F}$ is not merely hard to learn: every failed intervention degrades it first, a sensitive probe of whether hard-won structure survives.

If late-budget training is wasted and competence is bottlenecked on specific fragile structures, then a candidate intervention should be judged by two questions: *where* in training does it inject its signal, and *what* does it risk destroying? These two questions organize our search (§4.1) and motivate our method (§3.3).

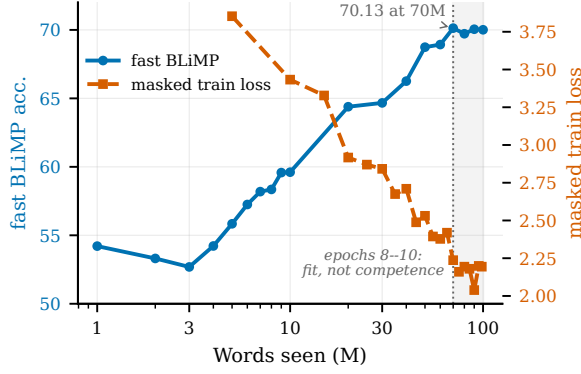


Figure 2: **The budget saturates early.** Fast BLiMP accuracy (left axis) reaches 70.13 by 70M words seen and is flat from there, while the masked training loss (right axis) keeps falling to the end of training. The final epochs lower the loss without improving competence.

3.3 Reallocating the Wasted Budget with MPO

The diagnosis licenses three design principles. The first is *timing*: refinement signal should arrive only after acquisition has saturated, where it no longer competes with the language-modeling gradient for capacity (§4.1 tests the necessity of this). The second is *preservation*: additions must be bounded in how far they can move the model, because $\mathcal{F}$ is what a strong new gradient destroys first. The third is *alignment*: the signal should share the evaluation’s functional form, which for BLiMP, its supplement, EWoK, and entity tracking is a relative-likelihood comparison between minimally different strings. Minimal-Pair Preference Optimization (MPO) instantiates all three (Figure 3).

Preference data comes from the corpus itself: no teacher, no reward model, no annotator, and no text beyond the ten million words already budgeted. A real sentence x^+ becomes $x^- = T(x^+)$ with T drawn uniformly from four edit operators, each length-preserving ($|T(x)| = |x|$): an adjacent-token swap, a distant-token swap, a function-word substitution, and a random content-token replacement. On *the cat sat on the mat* these produce, respectively, *the cat on sat the mat*, *the mat sat on the cat*, *the cat sat of the mat*, and *the cat lamp on the mat*. The four target complementary structure, from local word order to exactly the long-range configuration $\mathcal{F}$ tests: the adjacent swap probes local agreement, the distant swap and the function-word substitution probe the long-range dependencies of $\mathcal{F}$, and the random replacement supplies an easily rejected contrast that anchors the low end

of the difficulty range. Appendix E ablates each operator alone and confirms the complementarity. The phase touches under 3% of the 834,593 corpus sentence units, derives nothing from any evaluation resource, and introduces no new word types; the corruptions are input noise of the same kind the masked objective has injected since the first step of pretraining, which already replaces sampled tokens at random.

Length matching makes the comparison fair by construction. Every edit operates in place on the tokenized sentence, so x^+ and x^- contain the same number of tokens and their summed causal log-likelihoods,

$$\ell_\theta(x) = \sum_t \log \pi_\theta(x_t | x_{<t}), \quad (3)$$

are directly comparable with no normalization to tune, mirroring the minimal-pair design of BLiMP itself. (The suite scores the hybrid through a masked backend, §4.3; what the phase shares with the evaluation is this comparison structure, not the scoring function.) Given pairs, we optimize the DPO objective (cf. Rafailov et al., 2023):

$$\mathcal{L}_{\text{MPO}}(\theta) = -\mathbb{E} \left[\log \sigma \left(\beta [\Delta_\theta(x^+) - \Delta_\theta(x^-)] \right) \right], \quad (4)$$

with $\Delta_\theta(x) = \ell_\theta(x) - \ell_{\text{ref}}(x)$. The reference is a frozen copy of the checkpoint the policy starts from, and in the reward-maximization problem that Eq. 4 solves implicitly, β is the coefficient on $\text{KL}(\pi_\theta \parallel \pi_{\text{ref}})$ (Rafailov et al., 2023). Preservation therefore enters the objective directly rather than being assumed: likelihoods may reorder so that real sentences win, but drift from existing competence is penalized at rate $\beta = 0.1$. This property is what selects DPO among pairwise objectives: an anchor-free ranking loss would leave preservation to the learning rate alone, and the nearest anchor-free relative in this paper, the contrastive arm of §4.1, is the study’s clearest failure. That arm also supplies the closest control: the same four corruptions, injected during acquisition without an anchor and under a loss free to reshape the model, cost seven BLiMP points. The corruption data was never the problem; what changed between failure and success is when the signal arrives and how tightly it is held. Because preference optimization alone slowly erodes general language-modeling quality, a failure mode documented in the RLHF literature, the step- t loss retains the pretraining signal,

$$\mathcal{L}_{\text{phase}}(\theta) = \mathcal{L}_{\text{MPO}}(\theta) + m_t \mathcal{L}(\theta), \quad m_t \sim \text{Bernoulli}(0.5), \quad (5)$$

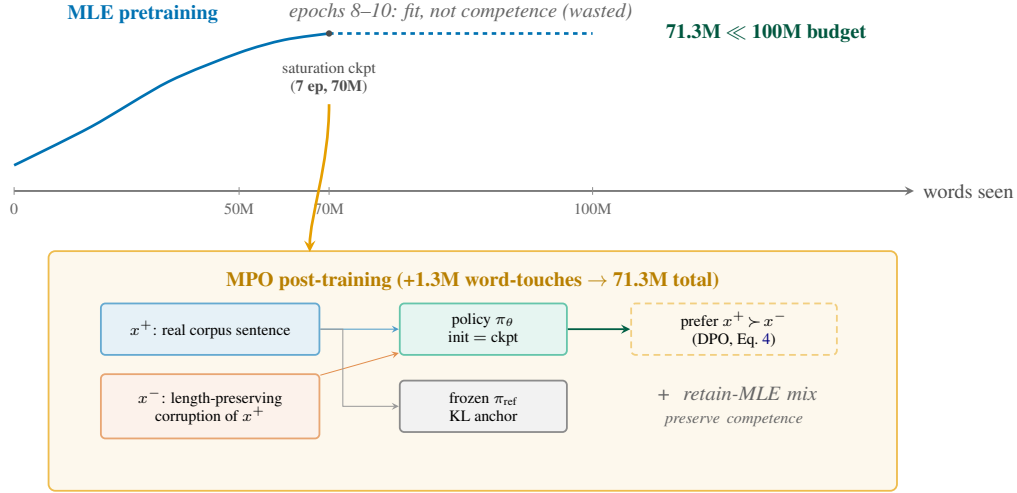


Figure 3: **Reallocating the wasted budget.** MLE pretraining saturates at the **7-epoch / 70M-word** checkpoint (dashed tail = the wasted epochs of Fig. 2). We fork there into **MPO**: the model is taught to prefer a real corpus sentence x^+ over a length-preserving corruption x^- of itself, KL-anchored to a frozen copy of the same checkpoint and mixed with a retain-MLE term (§3.3). Total exposure is 71.3M words, well under the 100M budget, and fewer than the fully-trained baseline.

one ordinary hybrid micro-batch (Eq. 1, post-anneal masking rate) at unit weight.

The timing principle fixes the starting point. Policy and reference initialize from the 70M-word saturation checkpoint, not the fully-trained model; running the identical phase from the 99M-word checkpoint reaches the same BLiMP (§4.2) at 29M additional words of budget, direct evidence that epochs eight through ten contributed nothing the method requires. Starting earlier does not substitute for acquisition: from the pre-saturation 30M checkpoint the same phase leaves fast BLiMP at 64.96, barely above that checkpoint’s own 64.67. The phase refines competence; it does not create it. Total exposure is 70M plus roughly 1.3M phase word-touches, counted conservatively: every token of both members of every pair and of the retained hybrid batches is charged against the budget, for 71.3M in all, well inside the 100M cap a 10-epoch run would exhaust (Appendix D itemizes the accounting).

The configuration is deliberately conservative: 1,500 AdamW steps at learning rate 10^{-5} , 16 pairs per step, cosine decay, clipping at 1.0, about 22 minutes on the same laptop (roughly 3% of the pre-training wall-clock), checkpoints every 250 steps. §4.2 shows a plateau across those checkpoints rather than a single favorable snapshot, and shows that a stronger preference objective improves its own training metrics while degrading the evaluation; the restraint is functional rather than stylistic.

4 Results

4.1 A Controlled Study of In-Training Interventions

The principles of §3.3 make falsifiable predictions: if timing binds, no intervention injected *during* acquisition should surpass the baseline; if preservation binds, unanchored signals should damage $\mathcal{F}$ before anything else; and if configuration sensitivity is as severe as the alignment principle assumes, gains imported from neighboring setups should not transfer. We test all three with a controlled study of seven in-training interventions spanning the sites where signal enters the update rule (objective, optimizer, weights, masking policy, token-level loss, tokenizer), each run holding budget, data order, and seed fixed while changing one component, and each gated by the same fast BLiMP-and-supplement evaluation. An eighth planned intervention, ensemble self-distillation, was excluded by hardware constraints (Table 2). The scope is deliberate: the predictions concern how the update rule reshapes the model, so the study varies that rule while holding the data stream fixed; data-side methods such as curricula lie outside it, and under a corpus traversed ten times, replay is the default rather than a distinct option. Table 2 collects the outcomes, and they fall where the principles predict. No in-training intervention improves on the baseline.

The objective-side interventions regress most

Model	NLP	BLiMP	Supp.	EWoK	Entity	PIQA	COMPS	GLUE
<i>Official baselines</i>								
GPT-2 (BabyLM 2026)	48.99	65.23	57.25	50.63	19.10	35.09	51.81	63.80
Interaction (BabyLM 2026)	49.04	63.07	58.13	51.45	19.62	36.14	51.29	63.62
<i>Other Strict-Small submissions (task-complete, ranked above ours)</i>								
BabySteps-MurphysLaw (rank 1)	53.59	71.60	63.93	51.94	27.95	36.15	53.13	70.44
www_curriculum (rank 2)	52.97	67.2	56.01	56.07	28.45	39.67	53.57	69.79
RecGPT-10M (rank 3)	52.40	73.11	61.73	52.62	16.59	40.68	55.43	66.64
<i>Ours</i>								
GPT-BERT (10 ep)	51.61	69.83	63.77	52.18	21.06	33.69	52.23	68.50
+ MPO (7.2 ep)	51.91	69.85	64.84	51.33	21.79	34.69	51.80	69.03

Table 1: **Main results on the official BabyLM 2026 Strict-Small metrics** (Overall NLP average and its seven task components; AoA and reading omitted, since AoA rounds to zero for nearly every submission under the corrected scorer, the GPT-2 baseline at -12.15 being the exception). Entity Tracking is the bias-filtered version; PIQA is GlobalPIQA. Comparison rows are the task-complete entries ranked above ours on the live leaderboard (fetched 2026-07-21, 07:55 UTC, 116 entries; submissions that have not rerun the new Entity/PIQA tasks are omitted). Our entries rank fourth and eighth; **bold** marks the best value per column among the tabulated entries. MPO improves over our fully-trained baseline on five of seven columns while using 7.2 rather than 10 epochs.

Intervention	Site	BLiMP	Suppl.
Baseline hybrid	—	70.01	65.20
Contrastive grammat. [†]	objective	62.87	64.80
Span masking	objective	68.61	60.00
Muon optimizer	optimizer	65.12	61.20
Checkpoint averaging	weights	69.96	65.20
Adaptive masking	masking	70.04	63.60
Selective LM	token loss	69.26	58.00
Morph.-aware tokenization	tokenizer	68.72	64.00
MPO (§3.3)	post-training	70.24	65.60*

Table 2: **Eight interventions, one improvement.** Fast BLiMP and BLiMP Supplement under an identical budget, recipe, and seed; every intervention changes one component relative to the baseline hybrid. [†]Contrastive training crashed at 77M words and is compared at the matched 70M checkpoint (reference 70.13/65.60). *Supplement varies with seed on this 50-item-per-subtask fast set (§4.2). A planned self-distillation intervention never produced a comparable run: three resident fp32 teachers exceed 16 GB, which we report as a hardware exclusion rather than a result.

severely. The contrastive grammaticality loss costs seven BLiMP points while its own auxiliary loss collapses toward zero, a shortcut at the task’s expense; span masking, a harder reconstruction task per epoch, underfits the fixed budget. Muon degrades both, a higher training loss than AdamW at equal words seen and weaker transfer besides, a mismatch with this budget-bound regime rather than a verdict on the optimizer. Averaging late checkpoints has no effect; by 80M words they occupy a shared basin. Adaptive masking arrives with a reported one-point BLiMP gain on a DeBERTa

encoder (Edman and Fraser, 2025) and transfers to our hybrid as a tie. Morphology-aware tokenization arrives reporting EWoK gains of roughly twenty percent and entity-tracking gains of roughly forty percent at this budget (Bölücü and Can, 2025) and inverts under our recipe: EWoK falls below chance, to 47.73. Selective language modeling (Lin et al., 2024; Mindermann et al., 2022) holds BLiMP but gives up seven supplement points: excess-loss selection de-weights the most predictable text, child-directed dialogue, and with it the pragmatics the supplement measures.

The regressions also share a structure, the preservation prediction made visible. In Figure 4, four mechanistically unrelated interventions degrade the same two filler-gap phenomena first, by 15 to 60 points, while typically gaining on easier local sub-tasks. Long-distance dependencies are both the last structure acquired and the first sacrificed when the signal is perturbed; a viable intervention must leave them alone. Nor does greater intensity rescue a proxy, which is why preservation requires an explicit anchor: the contrastive loss reaches near zero as BLiMP falls, Muon fits the MLM objective harder and generalizes worse, and §4.2 shows the same trade-off within our own method. At 33M parameters and 10M words there is no spare capacity, and a proxy pursued past the anchor’s tolerance is paid in competence. Portability fares no better: the masking-policy result does not survive the change from encoder to hybrid, and the tokenizer result inverts under a different recipe and evaluation pipeline. At this scale results are sensitive to

Init	Negatives	β	Seed	BLiMP	Suppl.
<i>no preference phase (baseline)</i>				70.01	65.20
70M	mixed	0.1	42	70.24	65.60
70M	mixed	0.1	43	70.14	64.40
70M	mixed	0.1	44	70.28	63.60
30M	mixed	0.1	42	64.96	62.80
99M	mixed	0.1	42	70.26	62.40
70M	hard	0.1	42	70.16	64.40
70M	hard	0.2	42	70.14	64.00

Table 3: **MPO ablations** (fast BLiMP / Supplement at the step-1250 operating point). Seeds vary the entire phase, not just data order. The two init rows bracket the saturation checkpoint: from 99M the phase matches the 70M-init BLiMP while spending 29M more words of budget, and from the pre-saturation 30M checkpoint it stays near that checkpoint’s own baseline (64.67/62.00), five points below every saturated init. “Hard” reweights corruptions toward the more adversarial edit types; together with the larger β it raises preference accuracy (0.85→0.90) and lowers evaluation scores. The supplement column is measured on 50 items per subtask and fluctuates accordingly.

interactions among architecture, recipe, and metric; re-validation under one’s own budget is a prerequisite. The study thus confirms each prediction: the region of the design space the principles exclude, in-training injection of unanchored signal, is empty as far as this search reaches, and the region they license contains MPO. Appendix F gives a mechanistic reading of the shared pattern in terms of basin geometry and gradient competition.

4.2 MPO

Table 1 reports both submitted models on the official Strict-Small metrics. MPO improves on the fully-trained baseline in five of seven task columns, gives back under a point on the other two (EWoK and COMPS), ties reading (0.27 vs. 0.29, both near zero), and lifts the NLP average from 51.61 to 51.91 at 7.2 rather than 10 epochs (71.3M vs. 99M words seen). The two entries place fourth and eighth among 116,¹ We sit near the top of the field on the BLiMP supplement; entity tracking and the world-knowledge tasks (EWoK, GlobalPIQA) are where we trail the leaders, a gap no measurement choice narrows. Across three seeds, fast BLiMP holds at 70.22 ± 0.07 and, uniquely among every intervention we tried, $\mathcal{F}$ improves (*wh_vs_that_with_gap* 71.5→75.5, con-

¹Model and all checkpoints are public at <https://huggingface.co/Ujjwal101/ujjwal-very-bored>. Leaderboard standings fetched 2026-07-21, 07:55 UTC.

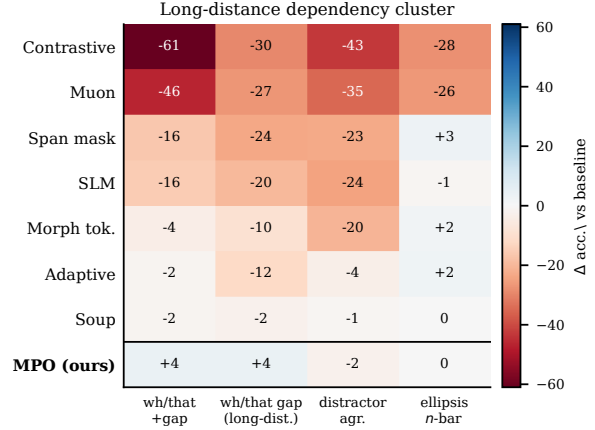


Figure 4: **Every failed intervention degrades the same cluster; only MPO repairs it.** Change in fast-BLiMP accuracy on the fragile long-distance dependency cluster $\mathcal{F}$ relative to the Phase-1 baseline, per intervention. Red = degraded, blue = improved. The objective- and optimizer-side interventions (Contrastive, Muon) reduce filler-gap licensing by 40–60 points; MPO (bottom) is the only intervention that *improves* it.

sistent across seeds and configurations); the full-suite gains on entity tracking, GlobalPIQA, and GLUE are from the submitted single-seed run (see Limitations).

The pattern across columns is inconsistent with teaching to the test. The benchmark nearest the training signal moves least: full BLiMP gains 0.02. The clear gains appear where the corruptions never reach, dialogue pragmatics in the supplement, entity tracking, GlobalPIQA, and finetuned GLUE, and Table 3 shows that pushing the proxy harder degrades the evaluation; a model exploiting the format would show the opposite, gains only where the format matches.

The direction of the small losses follows from what the signal contains. The corruptions manipulate structure, never facts, so the phase sharpens sensitivity to well-formedness but teaches nothing about the world; EWoK and COMPS contrast plausibility rather than grammar, and mass reallocated toward well-formedness leaves such contrasts untouched or marginally taxed. Both drops are under a point, and §3.2 shows world knowledge was never acquired in the first place: nothing there for refinement to refine.

Four alternative explanations could undermine the result; Table 3 collects the runs that test them. *Seed sensitivity.* Seeds 43 and 44 leave fast BLiMP at 70.22 ± 0.07 , the $\mathcal{F}$ repair recurs in every saturation-init run, and the 50-item supplement

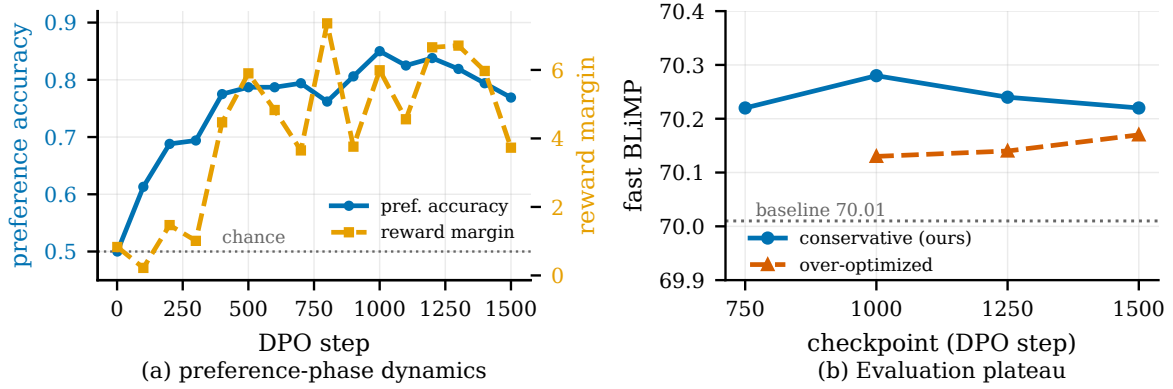


Figure 5: **MPO dynamics and the eval plateau.** (a) Preference accuracy rises to ~ 0.85 and the reward margin grows: the objective is learned. (b) Fast BLiMP across checkpoints: the conservative run (ours, $\beta=0.1$) holds a plateau above the baseline, ruling out checkpoint-selection effects, while the over-optimized variant ($\beta=0.2$, harder negatives) plateaus below it despite higher preference accuracy.

fluctuates as noise. *Checkpoint selection.* Figure 5b shows checkpoints 750 through 1,500 spanning 70.22 to 70.28, a plateau. *Initialization.* The starting point matters in both directions. From the 99M-word checkpoint the identical phase reaches 70.26, the same BLiMP for 29M more words; from the pre-saturation 30M checkpoint it reaches 64.96, a small gain over its own 64.67 and five points short of every saturated-init run. Refinement tracks its starting competence and substitutes for none of it; the diagnosis identifies a causal precondition, not a descriptive landmark. *Objective strength.* Hard negatives at $\beta = 0.1$ give 70.16; $\beta=0.2$ gives 70.14 while preference accuracy rises from 0.85 to 0.90, the proxy improving as the evaluation deteriorates, the §4.1 trade-off within our own method. The conservative configuration is not an untuned default but the optimum here.

4.3 Submission

Both models are separate leaderboard entries, the fully trained baseline and the MPO model; each is the organizers’ server-scored collation over the complete suite with the required checkpoints, so no number in Table 1 is from our own harness. The hybrid uses the pipeline’s masked next-token (mntp) backend for the zero-shot tasks and its causal backend for entity tracking and age of acquisition; the MPO entry declares its true 71.3M-word exposure against the 100M cap. The two entries hold fourth and eighth among 116; the board is live but the within-submission comparison is not, since the same 33.0M model appears twice, with and without the reallocated tail, every difference a 22-minute phase.

5 Conclusion

Under a fixed developmental budget, the cheapest resource is the one already being wasted. Our diagnosis found the final third of the Strict-Small epoch budget buying a lower training loss and no further competence, the two curves parting company exactly where the schedule keeps spending. That gap is the opening our method works through. Three principles, timing, preservation, and alignment, turn the observation into a recipe, and both a controlled study of seven interventions and the live leaderboard bear it out. The recipe is short: train until the evaluation saturates (Eq. 2), then spend what remains on anchored, evaluation-aligned refinement from that checkpoint, held close enough that the hard-won structure survives. Nothing in it is specific to our backbone, and both steps should carry to any model whose budget outlives its learning. We resist a strong developmental reading, though the shape is suggestive: exposure first, contrast after. At this scale, when the corpus is exhausted, the budget need not be.

Limitations

Our study is deliberately narrow in scope: a single 33.0M-parameter architecture, one English corpus, a single 10M-word budget. Whether the diagnosis and MPO carry to other backbones, larger budgets, or other languages remains to be shown, and the full-suite evaluation used one seed, though the two claims we rest on both held across three.

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Appendix

A Training Hyperparameters

Table 6 lists both phases end to end. The pretraining configuration stays deliberately close to the released GPT-BERT recipe, and the MPO phase adds no hyperparameter search beyond the ablations of §4.2. Every run in the paper, training, refinement, and evaluation alike, executed on the same laptop.

Two pretraining choices deserve comment. The 128-token sequence length is a hardware accommodation: the backbone’s relative-position attention leaves the fast Apple-silicon (MPS) execution path above 128 tokens, and the corpus, dominated by transcribed dialogue, loses little at this length. The effective batch of 128 sequences is reached through gradient accumulation (32×4), and the masking rate anneals from 0.30 to 0.15, so early epochs face a harder reconstruction problem than late ones.

The MPO phase runs at a learning rate fifty times below the pretraining peak, under a tighter gradient clip (1.0 against 2.0). Both choices serve the preservation principle of §3.3: the phase is meant to reorder likelihoods around an already competent solution, not to travel far from it. Checkpoints every 250 steps yield the trajectory of Figure 5b, from which step 1,250 is the operating point; the full phase costs 22 minutes against 12.5 hours of pretraining.

B Per-Benchmark Breakdowns

Phenomenon	10M	30M	50M	70M	99M
Anaphor number agr.	86.0	85.5	88.5	92.5	94.0
Det.-noun agr. (1)	80.5	86.0	88.0	90.5	90.0
Irreg. plural SV agr. (1)	69.5	71.5	72.5	74.5	73.5
Ellipsis <i>N</i> -bar (2)	53.5	83.0	91.0	91.5	91.5
Wh/that with gap	12.5	45.0	67.0	73.5	71.5
Distractor agr. (rel. noun)	22.5	25.0	32.0	57.5	62.0
Wh/that gap, long-dist.	0.5	10.0	26.0	34.0	32.5

Table 4: Acquisition timing on the baseline run: fast BLiMP subtask accuracy by words seen. Local morphology (top) is largely in place by 10M words; the long-distance cluster $\mathcal{F}$ (bottom) is acquired last and never fully.

Table 4 gives the acquisition side of the diagnosis. Local morphology is essentially in place by 10M words, a single epoch, and drifts upward from there. The long-distance cluster is different in kind: filler-gap licensing crosses 50% only after 30M words, its long-distance variant never reaches 35%, and

distractor agreement is still climbing when the budget ends, the one exception to the aggregate plateau. World knowledge remains at chance throughout: fast EWoK stays within 47–53 across all nineteen checkpoints and ends at 52.18 on the full set.

	wh gap	long-d.	distr.	ellipsis
Baseline	71.5	32.5	62.0	91.5
Contrastive obj.	10.5	3.0	19.0	63.5
Muon	25.0	5.5	27.0	65.0
Span masking	55.0	9.0	39.0	94.5
Selective LM	56.0	12.5	37.5	90.5
Morph. tokenization	67.5	23.0	41.5	93.5
Adaptive masking	70.0	20.5	58.5	94.0
Checkpoint avg.	70.0	30.0	61.0	91.5
MPO (ours)	75.5	36.5	59.5	91.5

Table 5: Absolute fast BLiMP accuracy on the fragile cluster $\mathcal{F}$ (the values behind Figure 4): wh gap = *wh_vs_that_with_gap*, long-d. = its long-distance variant, distr. = *distractor_agreement_relational_noun*, ellipsis = *ellipsis_n_bar_2*. MPO is the only intervention that improves the two filler-gap columns.

Two details of the table sharpen the picture. The hardest phenomena do not start at chance but below it: after one epoch the model prefers the corrupted member on *wh*-licensing items (12.5) and almost categorically on their long-distance variants (0.5), so acquisition here means unlearning a systematic early preference, not filling a gap in knowledge. Phenomena also differ in trajectory shape: *N*-bar ellipsis is acquired in a single burst between 10M and 30M words (53.5 to 83.0) and is stable thereafter, whereas distractor agreement climbs steadily to the end. The 70M column is the saturation point w^* of Eq. 2; the movement that remains past it is confined to this one phenomenon and gains too little to lift the aggregate.

Table 5 lists the absolute accuracies behind the deltas of Figure 4. All rows are final checkpoints except the contrastive grammaticality objective, which is measured at its 70M-word A/B point, where that run ended (§4.1).

Read column-wise, the table grades the interventions by how strongly they perturb the training signal. The objective-level change is the most destructive, leaving *wh*-licensing at 10.5, below even the one-epoch value of Table 4; the optimizer change follows at 25.0; the masking, token-selection, and tokenizer variants land between 55.0 and 67.5; and checkpoint averaging, which perturbs no gradients at all, sits at the baseline. *N*-bar ellipsis is the robust member of the cluster, degraded materi-

ally only by the two most destructive interventions (63.5 and 65.0 against 91.5). MPO’s repair is confined to the two filler–gap columns: its distractor-agreement value (59.5 against 62.0) is within the sampling noise of the 200-item fast sets, and even the repaired long-distance variant remains below chance. The cluster is improved, not solved.

C Saturation-Point Sensitivity

<i>Pretraining</i>	
Hidden / layers / heads / FFN	384 / 12 / 6 / 1,280
Parameters (tied embeddings)	33,046,084
Vocabulary (byte-level BPE)	16,384
Sequence length	128
Batch $\times$ grad. accum.	32 $\times$ 4 (eff. 128)
Optimizer	AdamW, lr 5×10^{-4}
(β_1, β_2) / weight decay	(0.9, 0.98) / 0.1
Schedule	cosine, 2% warmup
Gradient clip	2.0
Masked-mode share z	15/16 of micro-batches
Masking rate	0.30 $\rightarrow$ 0.15 (annealed)
Epochs / words seen	10 / $\approx$ 99M
Seed	42
Wall clock	12 h 27 m
<i>MPO phase</i>	
Init. / reference	70M ckpt (frozen copy)
β	0.1
Steps	1,500 (selected: 1,250)
Optimizer	AdamW, lr 10^{-5} , cosine
Pairs per step	16
Corruption mix	uniform over the four edits
Retained-MLE mix	$p = 0.5$, one hybrid batch
Gradient clip	1.0
Checkpoint interval	250 steps
Seed	42
Wall clock	22 min

Table 6: Complete hyperparameters for both phases. All runs use one M1 laptop (16 GB, MPS). GLUE fine-tuning uses the official recipe at sequence length 128, batch 32 (16 for BoolQ and MultiRC), 10 epochs (30 for WSC), with pipeline defaults otherwise.

The tolerance ε enters the diagnosis only through Eq. 2, and the milestone grid is spaced at 10M words past the first epoch, so w^* moves in 10M quanta. Table 7 sweeps it. No later checkpoint exceeds the 70M value, so $w^* = 70\text{M}$ down to $\varepsilon = 0$; the first alternative selections appear at $\varepsilon = 1.20$ (60M) and $\varepsilon = 1.39$ (50M), thresholds seventeen and twenty times the ± 0.07 cross-seed deviation of §4.2. A tolerance set anywhere within an order of magnitude of the measurement noise selects the same checkpoint, and even the first mis-selection, 60M, lies past the acquisition of every phenomenon in Table 4 except distractor agreement.

Tolerance ε	w^* (Eq. 2)
[0, 1.20)	70M
[1.20, 1.39)	60M
[1.39, 3.87)	50M
[3.87, 5.46)	40M

Table 7: Sensitivity of the saturation point to the tolerance. The selection used throughout the paper (70M) holds for every ε below 1.20 fast-BLiMP points.

D Exposure Accounting

Table 8 itemizes every token the submitted MPO model saw. Both members of every preference pair are charged in full, although x^- is an edit of an already counted corpus sentence, and the retained hybrid micro-batches (Eq. 5) are charged at their full sequence length. Component rows are expectations under the sampling distribution (24,000 pairs at the bank’s mean sentence length of 13.2 tokens; a Bernoulli(0.5) retain draw per step); the phase total is the realized value logged by the training loop. Tokens convert to words at the corpus ratio of 0.613. No text outside the ten-million-word corpus enters at any point.

Component	Basis	Words
MLE pretraining to w^*	7 epochs, logged	70.0M
Preference pairs (x^+, x^-)	48,000 sentences	$\approx$ 0.39M
Retained hybrid batches	$\approx$ 750 batches	$\approx$ 0.94M
Phase total (realized, logged)	training loop	1.28M
Total exposure		71.3M
Words-seen cap (10 epochs)		100M
Unique-text budget		10M

Table 8: Exposure accounting for the submitted MPO model. Total exposure is 71.3M of the 100M words-seen cap; the unique-text budget is untouched beyond the official corpus.

E Corruption Operators and Per-Operator Ablation

The four corruption operators behind MPO’s preference pairs act on the tokenized sentence, preserve its length, and never touch the start-of-sequence position or padding. The adjacent swap exchanges a random pair of neighboring tokens. The distant swap exchanges two positions drawn independently from the whole sentence. The function-word substitution replaces one closed-class token with a different one drawn from the corpus’s 120 most frequent non-special tokens, which in English are overwhelmingly determiners, prepositions, auxiliaries, and pronouns. The random replacement sub-

stitutes a uniformly drawn vocabulary token (§3.3 gives worked examples of all four). MPO samples the four uniformly; the hard-negative ablation reweights toward swaps and closed-class substitutions and away from the trivially detectable random replacement.

Table 9 reruns the identical phase with each operator alone (70M init, $\beta = 0.1$, seed 42, step-1250 canary). Three regularities emerge. First, aggregate transfer tracks the operator’s proximity to the evaluation’s own contrast type: the closed-class substitution, the edit nearest BLiMP’s morphosyntactic minimal pairs, is the strongest single choice; the swaps sit near the mixture; and the random replacement, the edit MPO already treats as trivially detectable, transfers worst. Second, the filler-gap repair follows a different ordering: the order-perturbing swaps repair *wh_vs_that_with_gap* most strongly (79.0 against the baseline’s 71.5), the substitution repairs it partially, and the random replacement not at all. The operators are complementary rather than redundant, as §3.3 intends. Third, every variant reaches the same preference accuracy (0.85–0.86): the objective is equally learnable from any operator, and the differences lie entirely in transfer, the proxy-evaluation dissociation of §4 in a third form. No single operator dominates: the substitution wins the aggregate, the swaps win the cluster repair, and only the mixture holds the supplement’s best value. The submitted model keeps the untuned uniform mixture.

Operator	BLiMP	Suppl.	wh gap	long-d.	p-acc
Adjacent swap	69.85	64.00	79.0	37.5	0.86
Distant swap	70.08	64.80	79.0	37.5	0.85
Function-word sub.	70.54	65.20	75.0	35.5	0.86
Random replace	69.83	64.00	72.5	32.5	0.86
Mixture (submitted)	70.24	65.60	75.5	36.5	0.85
No phase (baseline)	70.01	65.20	71.5	32.5	—

Table 9: Per-operator ablation: the MPO phase run with a single corruption operator, all else identical to the submitted configuration. Columns as in Table 5; **bold** marks the best value per column.

F A Mechanistic Reading

A mechanistic reading consistent with the outcomes of §4.1 and §4.2 is available. By 80M words the late checkpoints share a basin (the checkpoint-averaging result of Table 2), so the saturated model occupies a region of near-equivalent solutions; a small-learning-rate phase held to a frozen reference

moves within that region, reordering likelihoods without leaving it, the same role anchoring plays in continual learning (Kirkpatrick et al., 2017). An unanchored signal injected during acquisition instead competes with the language-modeling gradient for the same limited capacity, and what it sacrifices first is whatever that gradient holds most tenuously, the long-distance cluster. We offer this as an organizing account rather than a demonstrated mechanism; the observations it organizes are Tables 2 and 3.